Lecture 6: Particle Swarm Optimization (PSO)

Course: Evolutionary Algorithms

Instructor: Engr. Muhammad Farooq

Learning outcomes

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- update personal best and global best positions
- perform two PSO iterations for a small Rosenbrock example
- explain why boundary handling is needed in PSO

Lecture outline

1. Introduction to PSO

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7. First iteration: detailed calculation

Lecture outline

1. Introduction to PSO
2. Position and velocity of a particle
3. Personal best and global best positions
4. Velocity components
5. PSO algorithm and flow
6. Worked example using Rosenbrock function
7. First iteration: detailed calculation
8. Second iteration: detailed calculation

Lecture outline

Motivation: from swarm behavior to optimization

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- A swarm searches for food in a cooperative way.
- Each member learns from its own experience and from the group.

In optimization, each member of the swarm represents a possible solution. The swarm moves through the search space to locate the best solution.

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- It uses simple update equations instead of crossover and mutation.

Key idea: particles move in the search space by combining their previous direction, their own best experience, and the swarm's best experience.

PSO terminology

Term	Meaning
Swarm	Population of candidate solutions
Particle	One candidate solution in the swarm
Position	Current values of the decision variables
Velocity	Step used to move the particle
Personal best	Best position found so far by that particle
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For a two-variable problem, a particle position may be written as

$$\boldsymbol{x}_i = (x_{i,1}, x_{i,2})^T.$$

GA versus PSO: basic difference

Aspect	GA / RGA	PSO
Population member	individual / chromosome	particle
Main movement	crossover and mutation	velocity and position update
Memory	usually population-level	each particle keeps personal best
Group knowledge	selection pressure	global best attracts particles
Typical variables	binary or real-valued	usually real-valued

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PSO does not create offspring in the same way as GA. Instead, each particle moves to a new position.

Three distinct features of PSO

Personal best

Best position found by each particle

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Global best

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Velocity update

Rule used to move each particle

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Personal best	Global best	Velocity update
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Best position found by each particle	Best position found by the whole swarm	Rule used to move each particle

- Personal best preserves the particle's own memory.
- Global best shares the experience of the entire swarm.
- Velocity combines memory, self-experience, and social learning.

Position update

For particle i, the position is updated as:

$$\boldsymbol{x}_i^{(t+1)} = \boldsymbol{x}_i^{(t)} + \boldsymbol{v}_i^{(t+1)}$$

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- The velocity acts like a movement vector.

If a particle is at $\boldsymbol{x} = (2, 3)^T$ and velocity is $\boldsymbol{v} = (-1, 0.5)^T$, then the new position is $\boldsymbol{x} = (1, 3.5)^T$.

Velocity update equation

The velocity of particle i is updated using:

$$\boldsymbol{v}_i^{(t+1)} = w\boldsymbol{v}_i^{(t)} + c_1 r_1 \left(\boldsymbol{p}_{i,lb}^{(t)} - \boldsymbol{x}_i^{(t)}\right) + c_2 r_2 \left(\boldsymbol{p}_{gb}^{(t)} - \boldsymbol{x}_i^{(t)}\right)$$

- w is the inertia weight.

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Velocity components

$$\mathbf{v}_i^{(t+1)} = \underbrace{w\mathbf{v}_i^{(t)}}_{\text{inertia}} + \underbrace{c_1 r_1(\mathbf{p}_{i,lb}^{(t)} - \mathbf{x}_i^{(t)})}_{\text{cognitive}} + \underbrace{c_2 r_2(\mathbf{p}_{gb}^{(t)} - \mathbf{x}_i^{(t)})}_{\text{social}}$$

- **Inertia part:** remembers the previous movement direction.

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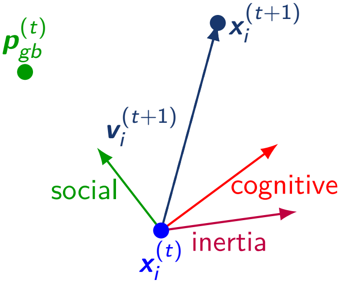
- **Inertia part:** remembers the previous movement direction.
- **Cognitive part:** pulls the particle toward its own best position.
- **Social part:** pulls the particle toward the swarm's best position.

The particle does not move randomly only. It moves by combining its previous direction, its own memory, and the swarm's best experience.

Geometrical view of velocity components



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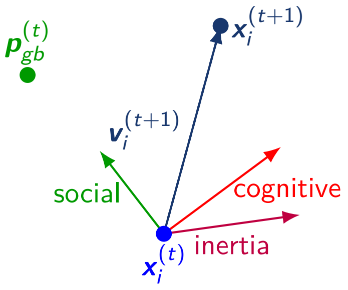

The final movement is the vector sum of the three components.

Personal best update

For a minimization problem, personal best is updated as:

$$\boldsymbol{p}_{i,lb}^{(t+1)} = \begin{cases} \boldsymbol{x}_i^{(t+1)}, & \text{if } f(\boldsymbol{x}_i^{(t+1)}) < f(\boldsymbol{p}_{i,lb}^{(t)}) \\ \boldsymbol{p}_{i,lb}^{(t)}, & \text{otherwise} \end{cases}$$

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- If the new position is better, update personal best.
- If the new position is worse, keep the previous personal best.

Global best update

For a minimization problem:

$$\boldsymbol{p}_{gb}^{(t)} = \text{best position among all personal best positions}$$

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The global best is the information shared with the whole swarm. It attracts all particles through the social component.

PSO algorithm

Basic PSO steps

1. Initialize particle positions and velocities.

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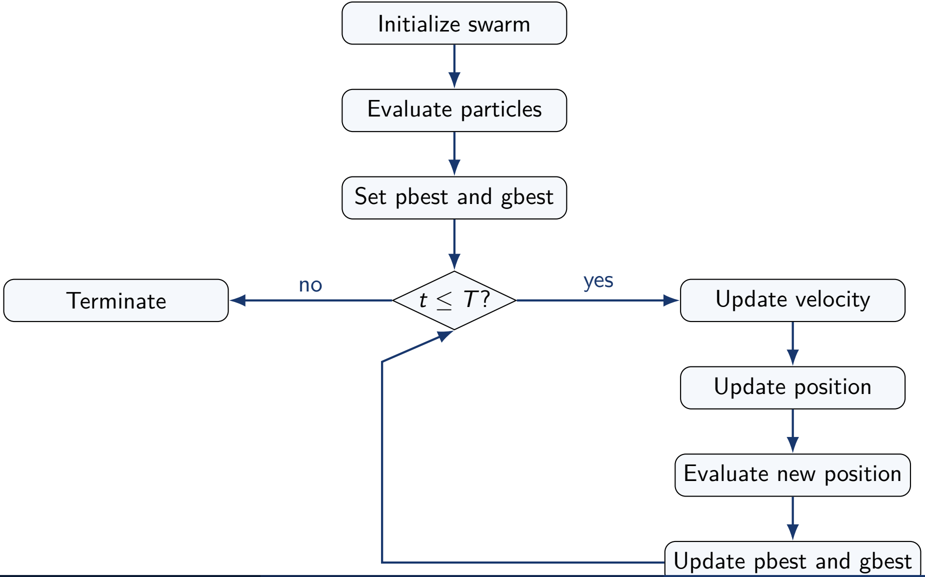
    - evaluate new positions
    - update personal best and global best
6. Stop when the maximum number of generations is reached.

PSO flowchart



Worked example: Rosenbrock function

We will minimize the Rosenbrock function:

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

with bounds

$$-5 \leq x_1 \leq 5, \qquad -5 \leq x_2 \leq 5.$$

- The global optimum is at $\boldsymbol{x}^* = (1, 1)^T$.

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- We will demonstrate PSO calculations for two iterations.

PSO parameter settings for the example

Parameter	**Value**
Number of variables	$n = 2$
Population size	$N = 8$
Initial generation	$t = 1$
Inertia weight	$w = 0.75$
Cognitive coefficient	$c_1 = 1.5$
Social coefficient	$c_2 = 2.0$
Initial velocity	$\boldsymbol{v}_i^{(1)} = (0, 0)^T$
Bounds	$[-5, 5]$ for both variables

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We assume minimization, so smaller function value means a better particle.

Initial swarm

Particle i	Position $\boldsymbol{x}_i^{(1)}$	Velocity $\boldsymbol{v}_i^{(1)}$
1	$(2.212,\ 3.009)^T$	$(0,0)^T$
2	$(-2.289,\ -2.396)^T$	$(0,0)^T$
3	$(-2.393,\ -4.790)^T$	$(0,0)^T$
4	$(-0.639,\ 1.692)^T$	$(0,0)^T$
5	$(-3.168,\ 0.706)^T$	$(0,0)^T$
6	$(0.215,\ -2.350)^T$	$(0,0)^T$
7	$(-0.742,\ 1.934)^T$	$(0,0)^T$
8	$(-4.563,\ 4.791)^T$	$(0,0)^T$

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At the start, all particles are placed randomly, and their velocities are set to zero for this example.

Evaluate the initial swarm

We calculate $f(x_1, x_2)$ for each particle.

Particle i	Position $\boldsymbol{x}_i^{(1)}$	$f(\boldsymbol{x}_i^{(1)})$
1	$(2.212,\ 3.009)^T$	357.154
2	$(-2.289,\ -2.396)^T$	5843.569
3	$(-2.393,\ -4.790)^T$	11066.800
4	$(-0.639,\ 1.692)^T$	167.414
5	$(-3.168,\ 0.706)^T$	8718.166
6	$(0.215,\ -2.350)^T$	574.796
7	$(-0.742,\ 1.934)^T$	194.618
8	$(-4.563,\ 4.791)^T$	25731.235

Initial personal best positions

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For example, for particle 1:

$$\boldsymbol{p}_{1,lb}^{(1)} = (2.212, 3.009)^T.$$

Initial global best position

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$$f(\boldsymbol{x}_4^{(1)}) = 167.414.$$

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Initial global best position

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Therefore, the global best is

$$\boldsymbol{p}_{gb}^{(1)} = \boldsymbol{x}_4^{(1)} = (-0.639, 1.692)^T.$$

This global best will attract all particles during the first velocity update.

First iteration: velocity update data

Use

$$\boldsymbol{v}_i^{(2)} = 0.75\boldsymbol{v}_i^{(1)} + 1.5r_1(\boldsymbol{p}_{i,lb}^{(1)} - \boldsymbol{x}_i^{(1)}) + 2.0r_2(\boldsymbol{p}_{gb}^{(1)} - \boldsymbol{x}_i^{(1)}).$$

Particle	r_1	r_2
1	0.661	0.312
2	0.919	0.271
3	0.782	0.824
4	0.299	0.055
5	0.874	0.595
6	0.133	0.582
7	0.031	0.736
8	0.366	0.954

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Because $\boldsymbol{p}_{i,lb}^{(1)} = \boldsymbol{x}_i^{(1)}$, the cognitive term is zero in the first velocity update.

First iteration: velocity of particle 1

For particle 1:

$$\boldsymbol{x}_1^{(1)} = (2.212, 3.009)^T, \quad \boldsymbol{v}_1^{(1)} = (0, 0)^T$$

$$\boldsymbol{p}_{1,lb}^{(1)} = (2.212, 3.009)^T, \quad \boldsymbol{p}_{gb}^{(1)} = (-0.639, 1.692)^T$$

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$$r_1 = 0.661, \qquad r_2 = 0.312.$$

For x_1 component:

$$v_{1,1}^{(2)} = 0.75(0) + 1.5(0.661)(2.212 - 2.212) + 2(0.312)(-0.639 - 2.212)$$

$$v_{1,1}^{(2)} = -1.779.$$

First iteration: velocity of particle 1

For particle 1:

$$\boldsymbol{x}_1^{(1)} = (2.212, 3.009)^T, \quad \boldsymbol{v}_1^{(1)} = (0, 0)^T$$

$$\boldsymbol{p}_{1,lb}^{(1)} = (2.212, 3.009)^T, \quad \boldsymbol{p}_{gb}^{(1)} = (-0.639, 1.692)^T$$

$$r_1 = 0.661, \qquad r_2 = 0.312.$$

For x_1 component:

$$v_{1,1}^{(2)} = 0.75(0) + 1.5(0.661)(2.212 - 2.212) + 2(0.312)(-0.639 - 2.212)$$

$$v_{1,1}^{(2)} = -1.779.$$

For x_2 component:

$$v_{1,2}^{(2)} = 0.75(0) + 1.5(0.661)(3.009 - 3.009) + 2(0.312)(1.692 - 3.009)$$

$$v_{1,2}^{(2)} = -0.822.$$

First iteration: velocity of particle 2

For particle 2:

$$\boldsymbol{x}_2^{(1)} = (-2.289, -2.396)^T, \quad \boldsymbol{p}_{2,lb}^{(1)} = (-2.289, -2.396)^T$$

$$\boldsymbol{p}_{gb}^{(1)} = (-0.639, 1.692)^T, \quad r_1 = 0.919, \quad r_2 = 0.271.$$

First iteration: velocity of particle 2

For particle 2:

$$\boldsymbol{x}_2^{(1)} = (-2.289, -2.396)^T, \quad \boldsymbol{p}_{2,lb}^{(1)} = (-2.289, -2.396)^T$$

$$\boldsymbol{p}_{gb}^{(1)} = (-0.639, 1.692)^T, \quad r_1 = 0.919, \quad r_2 = 0.271.$$

For x_1 component:

$$v_{2,1}^{(2)} = 0.75(0) + 1.5(0.919)(-2.289 + 2.289) + 2(0.271)(-0.639 + 2.289)$$

$$v_{2,1}^{(2)} = 0.893.$$

First iteration: velocity of particle 2

For particle 2:

$$\boldsymbol{x}_2^{(1)} = (-2.289, -2.396)^T, \quad \boldsymbol{p}_{2,lb}^{(1)} = (-2.289, -2.396)^T$$

$$\boldsymbol{p}_{gb}^{(1)} = (-0.639, 1.692)^T, \quad r_1 = 0.919, \quad r_2 = 0.271.$$

For x_1 component:

$$v_{2,1}^{(2)} = 0.75(0) + 1.5(0.919)(-2.289 + 2.289) + 2(0.271)(-0.639 + 2.289)$$

$$v_{2,1}^{(2)} = 0.893.$$

For x_2 component:

$$v_{2,2}^{(2)} = 0.75(0) + 1.5(0.919)(-2.396 + 2.396) + 2(0.271)(1.692 + 2.396)$$

$$v_{2,2}^{(2)} = 2.212.$$

First iteration: updated velocities

After applying the velocity update to all particles:

Particle	$v_{i,1}^{(2)}$	$v_{i,2}^{(2)}$
1	-1.779	-0.822
2	0.893	2.212
3	2.890	10.683
4	0.000	0.000
5	3.010	1.174
6	-0.994	4.704
7	0.151	-0.357
8	7.491	-5.916

First iteration: updated velocities

After applying the velocity update to all particles:

Particle	$v_{i,1}^{(2)}$	$v_{i,2}^{(2)}$
1	-1.779	-0.822
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6	-0.994	4.704
7	0.151	-0.357
8	7.491	-5.916

Large velocity values can push particles outside the search bounds, so boundary handling is needed after the position update.

First iteration: position update

Use

$$\boldsymbol{x}_i^{(2)} = \boldsymbol{x}_i^{(1)} + \boldsymbol{v}_i^{(2)}.$$

Particle	$\boldsymbol{x}_i^{(1)}$	$\boldsymbol{v}_i^{(2)}$	$\boldsymbol{x}_i^{(2)}$
1	$(2.212, 3.009)$	$(-1.779, -0.822)$	$(0.433, 2.187)$
2	$(-2.289, -2.396)$	$(0.893, 2.212)$	$(-1.396, -0.184)$
3	$(-2.393, -4.790)$	$(2.890, 10.683)$	$(0.498, 5.000)$
4	$(-0.639, 1.692)$	$(0.000, 0.000)$	$(-0.639, 1.692)$

First iteration: position update

Use

$$\boldsymbol{x}_i^{(2)} = \boldsymbol{x}_i^{(1)} + \boldsymbol{v}_i^{(2)}.$$

Particle	$\boldsymbol{x}_i^{(1)}$	$\boldsymbol{v}_i^{(2)}$	$\boldsymbol{x}_i^{(2)}$
1	$(2.212, 3.009)$	$(-1.779, -0.822)$	$(0.433, 2.187)$
2	$(-2.289, -2.396)$	$(0.893, 2.212)$	$(-1.396, -0.184)$
3	$(-2.393, -4.790)$	$(2.890, 10.683)$	$(0.498, 5.000)$
4	$(-0.639, 1.692)$	$(0.000, 0.000)$	$(-0.639, 1.692)$

For particle 3, the second component becomes 5.893, but the upper bound is 5. Therefore, we keep it at 5.000.

First iteration: position update continued

Particle	$\boldsymbol{x}_i^{(1)}$	$\boldsymbol{v}_i^{(2)}$	$\boldsymbol{x}_i^{(2)}$
5	$(-3.168, 0.706)$	$(3.010, 1.174)$	$(-0.157, 1.879)$
6	$(0.215, -2.350)$	$(-0.994, 4.704)$	$(-0.779, 2.354)$
7	$(-0.742, 1.934)$	$(0.151, -0.357)$	$(-0.590, 1.577)$
8	$(-4.563, 4.791)$	$(7.491, -5.916)$	$(2.928, -1.125)$

First iteration: position update continued

Particle	$\boldsymbol{x}_i^{(1)}$	$\boldsymbol{v}_i^{(2)}$	$\boldsymbol{x}_i^{(2)}$
5	$(-3.168, 0.706)$	$(3.010, 1.174)$	$(-0.157, 1.879)$
6	$(0.215, -2.350)$	$(-0.994, 4.704)$	$(-0.779, 2.354)$
7	$(-0.742, 1.934)$	$(0.151, -0.357)$	$(-0.590, 1.577)$
8	$(-4.563, 4.791)$	$(7.491, -5.916)$	$(2.928, -1.125)$

After updating positions, every particle is evaluated again using the objective function.

Evaluate swarm after first iteration

Particle	$\boldsymbol{x}_i^{(2)}$	$f(\boldsymbol{x}_i^{(2)})$
1	$(0.433, 2.187)^T$	399.984
2	$(-1.396, -0.184)^T$	460.648
3	$(0.498, 5.000)^T$	2258.514
4	$(-0.639, 1.692)^T$	167.414
5	$(-0.157, 1.879)^T$	345.375
6	$(-0.779, 2.354)^T$	308.580
7	$(-0.590, 1.577)^T$	153.484
8	$(2.928, -1.125)^T$	9406.994

Evaluate swarm after first iteration

Particle	$\boldsymbol{x}_i^{(2)}$	$f(\boldsymbol{x}_i^{(2)})$
1	$(0.433, 2.187)^T$	399.984
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4	$(-0.639, 1.692)^T$	167.414
5	$(-0.157, 1.879)^T$	345.375
6	$(-0.779, 2.354)^T$	308.580
7	$(-0.590, 1.577)^T$	153.484
8	$(2.928, -1.125)^T$	9406.994

The best particle after this position update is particle 7 with objective value 153.484.

Update personal bests after first iteration

For each particle, compare old and new objective values.

Particle	$f(\boldsymbol{x}_i^{(1)})$	$f(\boldsymbol{x}_i^{(2)})$	$\boldsymbol{p}_{i,lb}^{(2)}$
1	357.154	399.984	$(2.212, 3.009)^T$
2	5843.569	460.648	$(-1.396, -0.184)^T$
3	11066.800	2258.514	$(0.498, 5.000)^T$
4	167.414	167.414	$(-0.639, 1.692)^T$
5	8718.166	345.375	$(-0.157, 1.879)^T$
6	574.796	308.580	$(-0.779, 2.354)^T$
7	194.618	153.484	$(-0.590, 1.577)^T$
8	25731.235	9406.994	$(2.928, -1.125)^T$

Update global best after first iteration

- The best personal best after first iteration belongs to particle 7.

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$$\boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T.$$

Update global best after first iteration

- The best personal best after first iteration belongs to particle 7.
- Its objective value is 153.484.
- Therefore,

$$\boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T.$$

The global best has improved from particle 4 to particle 7.

Second iteration: velocity update data

Now use $\boldsymbol{x}_i^{(2)}$, $\boldsymbol{v}_i^{(2)}$, $\boldsymbol{p}_{i,lb}^{(2)}$, and $\boldsymbol{p}_{gb}^{(2)}$.

Particle	r_1	r_2
1	0.127	0.531
2	0.653	0.225
3	0.533	0.472
4	0.739	0.048
5	0.309	0.837
6	0.148	0.057
7	0.110	0.308
8	0.343	0.320

Second iteration: velocity update data

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8	0.343	0.320

Unlike the first update, the cognitive term is not always zero now because some personal bests are different from current positions.

Second iteration: particle 1 personal best

For particle 1:

$$\boldsymbol{x}_1^{(2)} = (0.433, 2.187)^T, \quad \boldsymbol{v}_1^{(2)} = (-1.779, -0.822)^T$$

$$f(\boldsymbol{x}_1^{(1)}) = 357.154 < f(\boldsymbol{x}_1^{(2)}) = 399.984.$$

Second iteration: particle 1 personal best

For particle 1:

$$\boldsymbol{x}_1^{(2)} = (0.433, 2.187)^T, \quad \boldsymbol{v}_1^{(2)} = (-1.779, -0.822)^T$$

$$f(\boldsymbol{x}_1^{(1)}) = 357.154 < f(\boldsymbol{x}_1^{(2)}) = 399.984.$$

Therefore, particle 1 keeps its previous personal best:

$$\boldsymbol{p}_{1,lb}^{(2)} = (2.212, 3.009)^T.$$

Second iteration: particle 1 personal best

For particle 1:

$$\boldsymbol{x}_1^{(2)} = (0.433, 2.187)^T, \quad \boldsymbol{v}_1^{(2)} = (-1.779, -0.822)^T$$

$$f(\boldsymbol{x}_1^{(1)}) = 357.154 < f(\boldsymbol{x}_1^{(2)}) = 399.984.$$

Therefore, particle 1 keeps its previous personal best:

$$\boldsymbol{p}_{1,lb}^{(2)} = (2.212, 3.009)^T.$$

Global best for the second velocity update is

$$\boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T.$$

Second iteration: velocity of particle 1

Use $r_1 = 0.127$ and $r_2 = 0.531$.
For x_1 component:

$$v_{1,1}^{(3)} = 0.75(-1.779) + 1.5(0.127)(2.212 - 0.433) \\ + 2(0.531)(-0.590 - 0.433)$$

$$v_{1,1}^{(3)} = -2.083.$$

Second iteration: velocity of particle 1

Use $r_1 = 0.127$ and $r_2 = 0.531$.
For x_1 component:

$$v_{1,1}^{(3)} = 0.75(-1.779) + 1.5(0.127)(2.212 - 0.433) + 2(0.531)(-0.590 - 0.433)$$

$$v_{1,1}^{(3)} = -2.083.$$

For x_2 component:

$$v_{1,2}^{(3)} = 0.75(-0.822) + 1.5(0.127)(3.009 - 2.187) + 2(0.531)(1.577 - 2.187)$$

$$v_{1,2}^{(3)} = -1.107.$$

Second iteration: velocity of particle 5

$$\boldsymbol{x}_5^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{v}_5^{(2)} = (3.010, 1.174)^T$$

$$\boldsymbol{p}_{5,lb}^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T$$

$$w = 0.75, \quad c_1 = 1.5, \quad c_2 = 2.0, \quad r_1 = 0.309, \quad r_2 = 0.837$$

Second iteration: velocity of particle 5

$$\boldsymbol{x}_5^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{v}_5^{(2)} = (3.010, 1.174)^T$$

$$\boldsymbol{p}_{5,lb}^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T$$

$$w = 0.75, \quad c_1 = 1.5, \quad c_2 = 2.0, \quad r_1 = 0.309, \quad r_2 = 0.837$$

Since $\boldsymbol{p}_{5,lb}^{(2)} = \boldsymbol{x}_5^{(2)}$, the cognitive term is zero.

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$$w = 0.75, \quad c_1 = 1.5, \quad c_2 = 2.0, \quad r_1 = 0.309, \quad r_2 = 0.837$$

Since $\boldsymbol{p}_{5,lb}^{(2)} = \boldsymbol{x}_5^{(2)}$, the cognitive term is zero.

$$v_{5,j}^{(3)} = 0.75 v_{5,j}^{(2)} + 2(0.837)\left(p_{gb,j}^{(2)} - x_{5,j}^{(2)}\right)$$

Second iteration: velocity of particle 5

$$\boldsymbol{x}_5^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{v}_5^{(2)} = (3.010, 1.174)^T$$

$$\boldsymbol{p}_{5,lb}^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T$$

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Since $\boldsymbol{p}_{5,lb}^{(2)} = \boldsymbol{x}_5^{(2)}$, the cognitive term is zero.

$$v_{5,j}^{(3)} = 0.75 v_{5,j}^{(2)} + 2(0.837) \left(p_{gb,j}^{(2)} - x_{5,j}^{(2)} \right)$$

Variable 1

$$v_{5,1}^{(3)} = 0.75(3.010) + 2(0.837)(-0.590 + 0.157)$$

$$= 2.2575 - 0.7248$$

$$v_{5,1}^{(3)} = 1.533$$

Second iteration: velocity of particle 5

$$\boldsymbol{x}_5^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{v}_5^{(2)} = (3.010, 1.174)^T$$

$$\boldsymbol{p}_{5,lb}^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T$$

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$$= 2.2575 - 0.7248$$

$$v_{5,1}^{(3)} = 1.533$$

Variable 2

$$v_{5,2}^{(3)} = 0.75(1.174) + 2(0.837)(1.577 - 1.879)$$

$$= 0.8805 - 0.5055$$

$$v_{5,2}^{(3)} = 0.375$$

Second iteration: velocity of particle 5

$$\boldsymbol{x}_5^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{v}_5^{(2)} = (3.010, 1.174)^T$$

$$\boldsymbol{p}_{5,lb}^{(2)} = (-0.157, 1.879)^T, \quad \boldsymbol{p}_{gb}^{(2)} = (-0.590, 1.577)^T$$

$$w = 0.75, \quad c_1 = 1.5, \quad c_2 = 2.0, \quad r_1 = 0.309, \quad r_2 = 0.837$$

Since $\boldsymbol{p}_{5,lb}^{(2)} = \boldsymbol{x}_5^{(2)}$, the cognitive term is zero.

$$v_{5,j}^{(3)} = 0.75 v_{5,j}^{(2)} + 2(0.837)\left(p_{gb,j}^{(2)} - x_{5,j}^{(2)}\right)$$

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$$v_{5,1}^{(3)} = 0.75(3.010) + 2(0.837)(-0.590 + 0.157)$$

$$= 2.2575 - 0.7248$$

$$v_{5,1}^{(3)} = 1.533$$

Variable 2

$$v_{5,2}^{(3)} = 0.75(1.174) + 2(0.837)(1.577 - 1.879)$$

$$= 0.8805 - 0.5055$$

$$v_{5,2}^{(3)} = 0.375$$

$$\therefore \quad \boldsymbol{v}_5^{(3)} = (1.533, 0.375)^T$$

Second iteration: updated velocities

After applying the velocity update to all particles:

Particle	$v_{i,1}^{(3)}$	$v_{i,2}^{(3)}$
1	-2.083	-1.107
2	1.033	2.452
3	1.140	3.934
4	0.005	-0.011
5	1.533	0.374
6	-0.724	3.439
7	0.114	-0.268
8	3.364	-2.705

Second iteration: position update

Use

$$\boldsymbol{x}_i^{(3)} = \boldsymbol{x}_i^{(2)} + \boldsymbol{v}_i^{(3)}.$$

Particle	$\boldsymbol{x}_i^{(3)}$ before bounds	$\boldsymbol{x}_i^{(3)}$ after bounds
1	$(-1.650, 1.080)^T$	$(-1.649, 1.079)^T$
2	$(-0.363, 2.268)^T$	$(-0.363, 2.268)^T$
3	$(1.638, 8.934)^T$	$(1.638, 5.000)^T$
4	$(-0.634, 1.681)^T$	$(-0.634, 1.681)^T$

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3	$(1.638, 8.934)^T$	$(1.638, 5.000)^T$
4	$(-0.634, 1.681)^T$	$(-0.634, 1.681)^T$

Particle 3 exceeds the upper bound in the second variable, so x_2 is limited to 5.000.

Second iteration: position update continued

Particle	$\boldsymbol{x}_i^{(3)}$ before bounds	$\boldsymbol{x}_i^{(3)}$ after bounds
5	$(1.376, 2.253)^T$	$(1.376, 2.254)^T$
6	$(-1.503, 5.793)^T$	$(-1.503, 5.000)^T$
7	$(-0.476, 1.309)^T$	$(-0.477, 1.309)^T$
8	$(6.292, -3.830)^T$	$(5.000, -3.830)^T$

Second iteration: position update continued

Particle	$\boldsymbol{x}_i^{(3)}$ before bounds	$\boldsymbol{x}_i^{(3)}$ after bounds
5	$(1.376, 2.253)^T$	$(1.376, 2.254)^T$
6	$(-1.503, 5.793)^T$	$(-1.503, 5.000)^T$
7	$(-0.476, 1.309)^T$	$(-0.477, 1.309)^T$
8	$(6.292, -3.830)^T$	$(5.000, -3.830)^T$

Particles 6 and 8 also exceed a bound, so their values are clipped to the nearest allowed boundary.

Evaluate swarm after second iteration

Particle	$\boldsymbol{x}_i^{(3)}$	$f(\boldsymbol{x}_i^{(3)})$
1	$(-1.649, 1.079)^T$	276.367
2	$(-0.363, 2.268)^T$	458.327
3	$(1.638, 5.000)^T$	537.876
4	$(-0.634, 1.681)^T$	166.098
5	$(1.376, 2.254)^T$	13.222
6	$(-1.503, 5.000)^T$	575.777
7	$(-0.477, 1.309)^T$	119.231
8	$(5.000, -3.830)^T$	83134.582

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Particle	$\boldsymbol{x}_i^{(3)}$	$f(\boldsymbol{x}_i^{(3)})$
1	$(-1.649, 1.079)^T$	276.367
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The best value after the second iteration is particle 5 with $f = 13.222$.

Update best positions after second iteration

- Some particles improve their personal best again.

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- Particle 5 improves significantly from 345.375 to 13.222.
- Particle 7 also improves from 153.484 to 119.231.
- The new global best is therefore particle 5:

$$\boldsymbol{p}_{gb}^{(3)} = (1.376, 2.254)^T, \qquad f(\boldsymbol{p}_{gb}^{(3)}) = 13.222.$$

Progress over two iterations

Stage	Global best position	Global best value
Initial swarm	$(-0.639, 1.692)^T$	167.414
After first iteration	$(-0.590, 1.577)^T$	153.484
After second iteration	$(1.376, 2.254)^T$	13.222

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Stage	Global best position	Global best value
Initial swarm	$(-0.639, 1.692)^T$	167.414
After first iteration	$(-0.590, 1.577)^T$	153.484
After second iteration	$(1.376, 2.254)^T$	13.222

The swarm has not reached the optimum yet, but the global best has improved strongly in only two iterations.

What students should notice from the example

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- In the first velocity update, the cognitive term is zero because $\boldsymbol{p}_{i,lb}^{(1)} = \boldsymbol{x}_i^{(1)}$.
- After particles move, personal best and global best may change.
- Once personal best differs from current position, the cognitive term becomes active.
- The social term pulls particles toward the global best.
- Some particles may move outside the variable bounds.
- Boundary handling keeps particles inside the allowed search space.

Common boundary handling methods

- **Clipping:** if a value exceeds the bound, set it to the nearest bound.

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- **Random reset:** replace the infeasible component with a random feasible value.

In this lecture example, we used clipping. For example, if $x_2 = 5.893$ and the upper bound is 5, we set $x_2 = 5$.

Discussion questions

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- Why is the cognitive term zero in the first velocity update of this example?
- Why do we need boundary handling in PSO?

Quick quiz

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- A. gene

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- B. particle

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- A. the best solution found by one particle
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Students should leave this lecture with these key points:

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- Boundary handling is needed when particles move outside limits.
- A complete PSO iteration includes velocity update, position update, evaluation, and best-position update.